

BIOLOGY 2010 June

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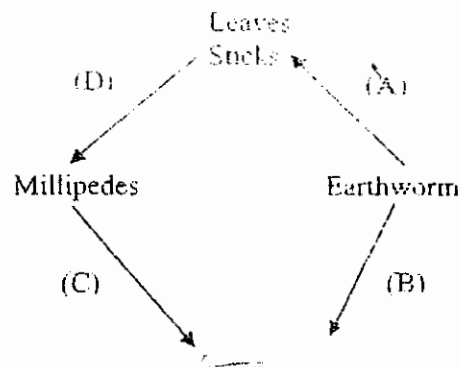
1. Some sharks have an excellent sense of smell, which allows them to detect tiny amounts of blood dissolved in water, and track down distant wounded animals. The shark can be described as

- I. predator
 - II. parasite
 - III. omnivore
 - IV. carnivore
- (A) I only
(B) I and IV only
(C) II and III only
(D) I, II, III and IV

2. Which of the following are involved in the cycling of nutrients in nature?

- I. Bacteria
 - II. Plants
 - III. Earthworms
 - IV. humans
- (A) I and II only
(B) III and IV only
(C) I, II and III only
(D) I, II, III and IV

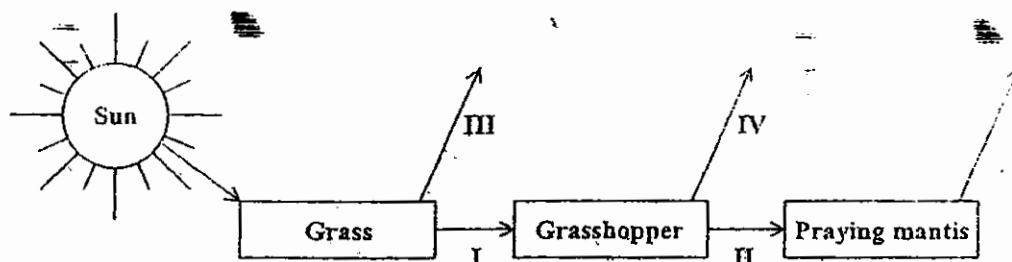
3. Which arrow in the food web diagram below points in the wrong direction?



4. Some insects are brightly coloured, but are difficult to spot as they sit on plants. This bright coloration may be important for

- I. pollination
 - II. blending into their environment
 - III. attracting mates
 - IV. hunting for prey
- (A) I and II only
(B) I and III only
(C) II, III, and IV only
(D) I, II, III and IV

Items 5 - 6 refer to the figure below which shows the flow of energy through a food chain.



5. Which arrow represents the smallest transfer of energy between organisms?

- (A) I
(B) II
(C) III
(D) IV

6. At which point is MOST of the energy released to the environment?

- (A) I
(B) II
(C) III
(D) IV

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Item 7 refers to the following terms.

- I. Organ
- II. Organism
- III. Organelle
- IV. Cell
- V. Tissue

7. Which of the following shows the correct sequence for INCREASING complexity/specialisation?

- (A) III, IV, I, II, V
- (B) III, IV, V, I, II
- (C) IV, I, II, III
- (D) IV, V, I, III, II

8. Which of the following processes is involved in the passage of water through cell walls?

- (A) Diffusion
- (B) Osmosis
- (C) Assimilation
- (D) Active transport

9. A mature plant cell is different from a mature animal cell because the plant cell has

- (A) mitochondria
- (B) a cell membrane
- (C) a large permanent vacuole
- (D) a nucleus suspended in the cytoplasm

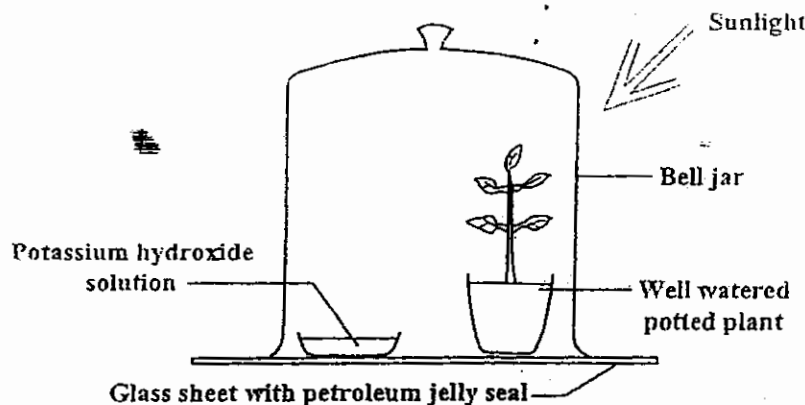
10. The process by which food is moved through the alimentary canal is called

- (A) digestion
- (B) mastication
- (C) peristalsis
- (D) swallowing

11. Which of the following are NOT autotrophs?

- (A) Algae
- (B) Mosses
- (C) Ferns
- (D) Fungi

Item 12 refers to the following diagram.



12. A well-watered plant is set up as shown in the diagram above and left for a few hours in sunlight. A leaf from the plant is then tested for starch and no blue-black colour appeared.

A likely explanation for this result is that the

- (A) soil in the pot dried out
- (B) jar contained no carbon dioxide
- (C) apparatus was not left long enough in sunlight
- (D) petroleum jelly prevented air from entering the bell jar

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Item 13 refers to the table below which shows the results obtained when a food was tested by adding different reagents.

Test	Result
Iodine	Blue-black colour
Sodium hydroxide and copper sulphate (Biuret reagent)	Purple colour
Benedict's or Fehling's solution and heating	No colour change

13. The food contained

- ☒ (A) starch and protein only
- ☐ (B) reducing sugar and starch only
- ☐ (C) reducing sugar, starch and protein
- ☐ (D) protein, fat and non-reducing sugar

14. Which of the following are essential for the formation of healthy bones?

- ☒ (A) Calcium and Vitamin D
- ☐ (B) Calcium and Vitamin E
- ☐ (C) Iron and Vitamin A
- ☐ (D) Iron and Vitamin C

15. Which of the following statements BEST describes the function of ATP?

- ☐ (A) It stores food.
- ☒ (B) It is a means of energy transfer.
- ☐ (C) It speeds up chemical reactions in the cell.
- ☐ (D) Large quantities of energy can become available.

16. Which of the following is NOT a reason for placing green plants in an aquarium?

- ☐ (A) Providing food for the animals
- ☐ (B) Providing oxygen for the animals
- ☒ (C) Removing the chlorine from the water
- ☐ (D) Removing the animals' waste products

17. Which of the following characteristics is NOT a feature of respiratory surfaces?

- ☐ (A) Large surface area
- ☐ (B) Permanently moist
- ☐ (C) Good supply of blood
- ☒ (D) Thick membrane

18. Which of the following statements about aerobic and anaerobic respiration is correct?

	Aerobic	Anaerobic
(A)	Does not use oxygen	Uses oxygen
(B)	Occurs in the cytoplasm	Occurs in the mitochondria
☒ (C)	Releases large amounts of energy	Releases small amounts of energy
☒ (D)	At least one product is organic	Products are always inorganic

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Item 19 refers to the following table which shows the results, recorded after 20 minutes, of tests done with cobalt chloride paper on two leaves from different plants.

% Colour change		
Leaf	Surface 1	Surface 2
X	100	10
Y	100	50

19. From the results, one may conclude that

- (A) Leaf X has more stomata on Surface 2 than Leaf Y.
- (B) Leaf Y has more stomata on Surface 1 than Leaf X.
- (C) Leaf X has a faster rate of transpiration than Leaf Y.
- (D) Leaf Y has a faster rate of transpiration than Leaf X.

20. As organisms increase in size, the process of diffusion limits the transport of substances because

- (A) their surface area to volume decreases
- (B) their surface area to volume increases
- (C) they develop transport systems
- (D) they produce more waste products

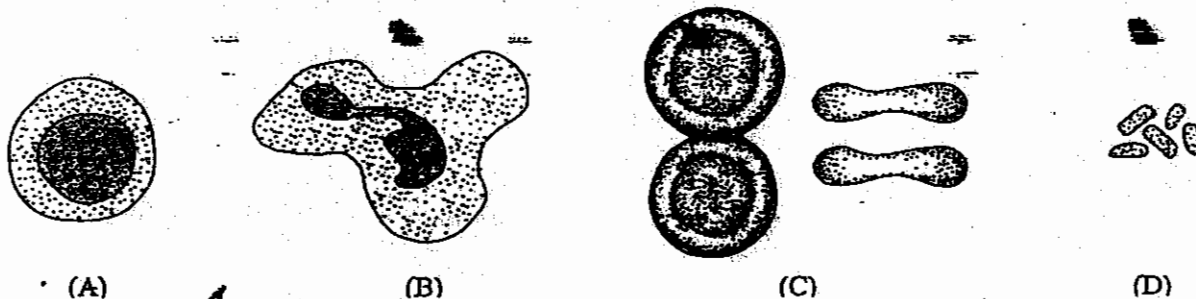
21. Which of the following plants does NOT have a modified vegetative storage organ?

- (A) Carrot
- (B) Ginger
- (C) Mango
- (D) Potato

22. Through which process does the body rid itself of nitrogenous waste products?

- (A) Excretion
- (B) Diffusion
- (C) Evaporation
- (D) Osmoregulation

Items 23 - 24 refer to the diagrams below which represent some components of blood.



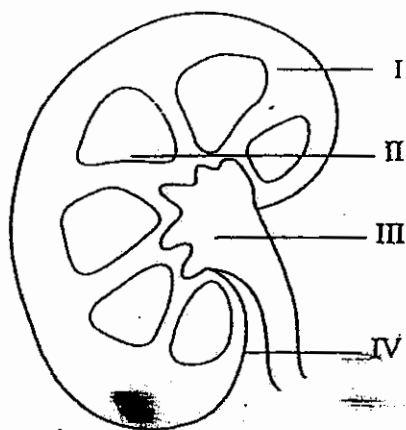
In answering items 23 - 24, match each item with one of the options above. Each option may be used once, more than once or not at all.

23. Engulfing disease-causing organisms

24. Transport of oxygen

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Items 25 - 26 refer to the diagram of the mammalian kidney.



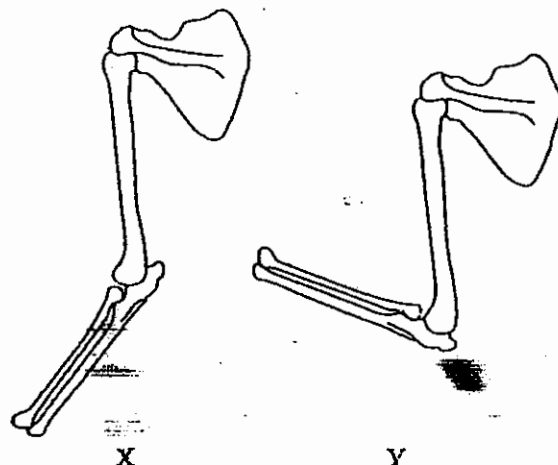
25. In which part of the kidney is the loop of Henle located?

- (A) I
- ☒ (B) II
- (C) III
- (D) IV

26. Into which structure is urine emptied?

- (A) I
- (B) II
- ☒ (C) III
- (D) IV

27. In the movement of the upper arm from position X to position Y in the diagram below, which of the following describes the action of biceps and triceps muscles?



- I. Biceps contract
- II. Biceps relax
- III. Triceps contract
- IV. Triceps relax

- (A) I and III only
- ☒ (B) I and IV only
- (C) II and III only
- (D) II and IV only

28. Which of the following adaptations in plants is NOT used to conserve water?

- (A) Thorns
- (B) Small leaves
- ☒ (C) Long leaves
- (D) Cuticle thickness

29. Movement is important to the survival of a plant because it ensures that it

- I. can find food
- II. grows towards light
- III. can find water
- IV. can reproduce

- (A) I only
- ☒ (B) I and IV only
- (C) II and III only
- (D) I, II, III and IV

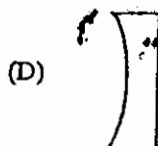
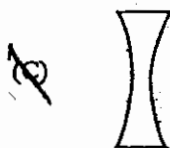
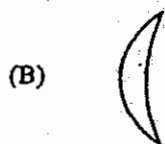
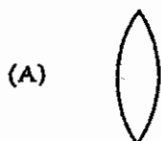
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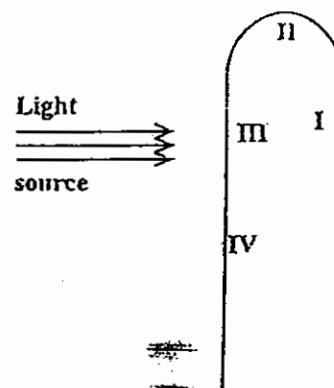
30. The human skeleton can be divided into two parts, the axial skeleton and the appendicular skeleton. Which row in the table groups bones correctly?

	Axial skeleton	Appendicular skeleton
(A)	Pelvic girdle, ribs, femur	Skull, humerus, vertebral column
(B)	Humerus, femur, ribs	Skull, vertebral column, pelvic girdle
(C)	Pelvic girdle, humerus, femur	Vertebral column, ribs, skull
(D)	Skull, vertebral column, ribs	Humerus, femur, pelvic girdle

31. Which of the following types of lens will correct shortsightedness?



Items 32 - 33 refer to the figure showing a diagram of the tip of a stem and a source of light.



32. After three days, where will the concentration of auxin be HIGHEST?

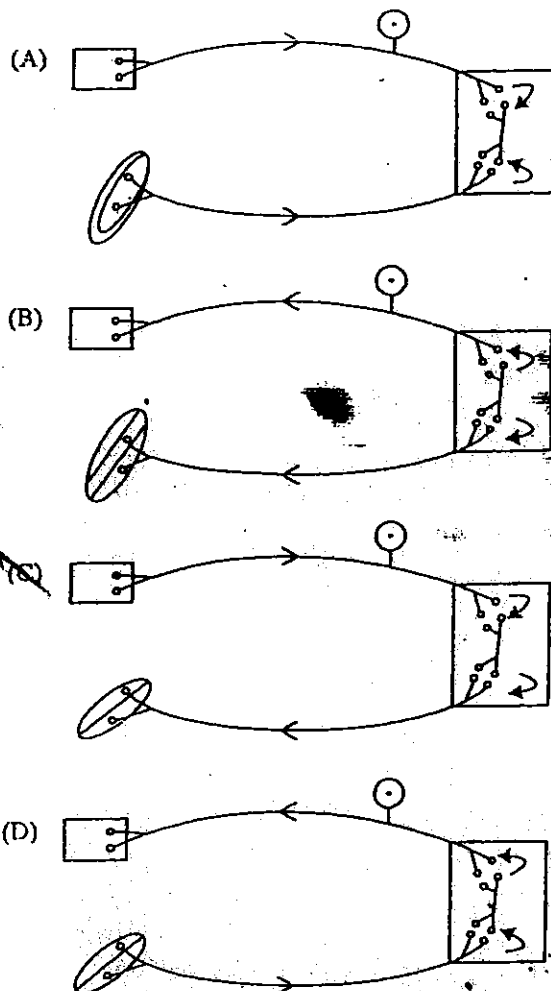
- ~~(A)~~ I
(B) II
(C) III
(D) IV

33. After three days, what will be the response of the stem?

- (A) Bend to the right
~~(B)~~ Bend to the left
(C) Increase in height
(D) No response

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34. Which of the following diagrams correctly represents the pathway of the transmission of an impulse for a spinal reflex?



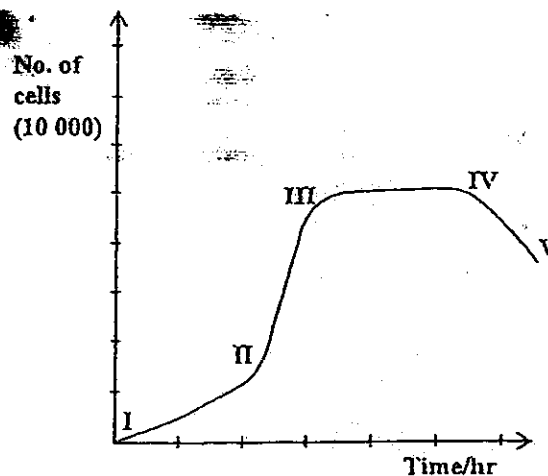
35. Which part of the brain would be used in solving a mathematical problem?

- (A) Cerebrum
(B) Cerebellum
(C) Hypothalamus
(D) Medulla oblongata

36. Which of the following functions of the skin is an example of homeostasis?

- (A) Storage of fats
(B) Release of sweat
(C) Secretion of sebum
(D) Maintenance of body temperature

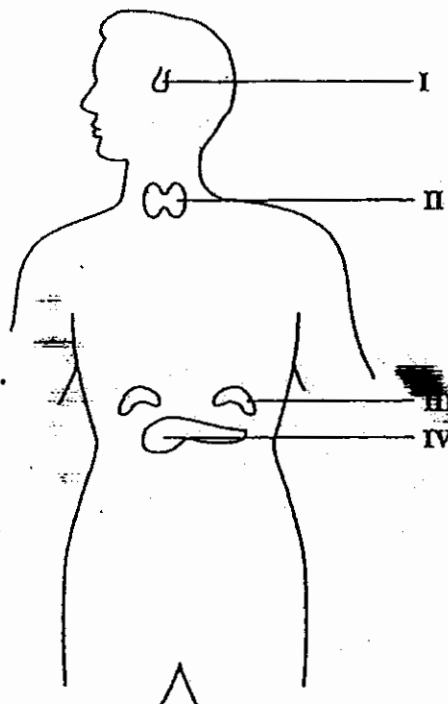
37. The diagram below represents the growth curve of a culture of yeast cells. Which section of the graph shows the MOST rapid rate of growth?



- (A) I - II
(B) II - III
(C) III - IV
(D) IV - V

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38. Which of the following glands produces a hormone which is directly responsible for growth in human beings?



- ☒ (A) I
(B) II
(C) III
(D) IV

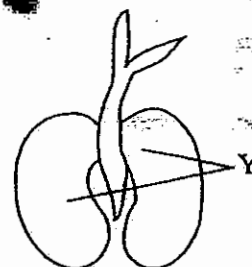
39. Which of the following sequences of events is correct?

- (A) Fertilization → implantation → copulation → gestation
(B) Fertilization → implantation → gestation → copulation
☒ (C) Copulation → fertilization → implantation → gestation
(D) Copulation → implantation → fertilization → gestation

40. The reproduction of amoeba by binary fission is an example of asexual reproduction because

- (A) the nucleus divides by mitosis
(B) the nucleus is not involved in the cell division
☒ (C) two offspring are produced by one parent
(D) the cytoplasm is divided into two equal halves

Items 41 - 42 refer to the diagram below.



41. The type of seed represented above is

- (A) monocotyledonous
☒ (B) dicotyledonous
(C) wind dispersed
(D) water dispersed

42. The MAIN food reserves found in the parts labelled Y are likely to be

- (A) protein and fats
(B) protein and calcium
(C) starch and fats
☒ (D) starch and protein

43. Which of the following statements is TRUE of wind dispersed seeds?

- ☒ (A) They have feathery hairs which allow them to be taken great distances.
(B) They have hooks which allow them to be taken great distances.
(C) They are rapidly strewn from dry pods.
(D) They are found in succulent fruit.

44. Which of the following events does NOT occur during mitosis?

- (A) Chromosome makes an exact copy of itself.
- (B) Chromosomes align themselves on the equator of the spindle.
- (C) Two chromatids are joined by a centromere.
- ☒ (D) The chromatids exchange genetic information.

45. The production of gametes during the process of meiosis can lead to

- (A) smaller population size
- (B) larger population size
- ☒ (C) variation in a population
- (D) no change in the characteristics of a population

46. Continuous variation in humans is illustrated by

- (A) sex/gender
- (B) tongue rolling
- (C) blood groups
- ☒ (D) adult heights

47. A characteristic is said to be sex linked if it

- (A) only affects one particular sex
- ☒ (B) is carried on the sex chromosomes
- (C) develops during sexual intercourse
- (D) is passed on from father to son

Item 48 refers to the table below which shows the numbers of green and brown grasshoppers found in equal areas of dry and green grass respectively.

	Brown grasshoppers	Green grasshoppers
Dried grass	19	6
Green grass	8	25

Which of the following is the BEST explanation for these observations?

- ☒ (A) Camouflaging protects grasshoppers from predators.
- (B) Brown grasshoppers mutate when placed in green areas.
- (C) Green grasshoppers are better adapted.
- (D) More food is available in green areas.

49. If a diploid organism has two different alleles for the same gene, it is described as being

- (A) dominant
- (B) homozygous
- ☒ (C) heterozygous
- (D) recessive

50. Which of the following statements about the differences between artificial and natural selection are true?

	Natural selection	Artificial selection
I.	Environment selects traits.	Humans select traits.
II.	Based on entire genetic make-up of individuals.	Based on specific genes in individuals.
III.	Organisms are well adapted to the environment.	Humans create the environment so that organisms can thrive.

- (A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

51. Which of the following correctly lists examples of pathogenic, nutritional deficiency and physiological diseases?

	Pathogenic	Nutritional deficiency	Physiological
(A)	Diabetes	AIDS	Anaemia
(B)	AIDS	Diabetes	Anaemia
(C)	Anaemia	Diabetes	AIDS
(D)	AIDS	Anaemia	Diabetes

52. In which of the ways listed below can hypertension be treated or controlled?

- I. Eating iron-rich foods
II. Low carbohydrate diet
III. Low salt intake
IV. Exercise
- (A) I and II only
(B) II and III only
(C) III and IV only
(D) I, II, III and IV

53. An animal which carries a disease-causing organism but is not adversely affected by it is known as a

- (A) host
(B) vector
(C) parasite
(D) pathogen

54. Which of the following symptoms indicates the abuse of alcohol?

- (A) Slowed muscular skills, increased body temperature, feelings of increased energy
(B) Vasodilation, impaired mental functioning, loss of consciousness
(C) Vasoconstriction, violent behaviour, reduced need for sleep
(D) Slurred speech, loss of appetite, lung and nasal damage

55. Which of the following statements are implications of diseases in agricultural crops?

- I. Loss of crops
II. More jobs for workers in the area
III. Lower income for government
IV. Higher food import cost

- (A) I and IV only
(B) II and III only
(C) I, III and IV only
(D) II, III and IV only

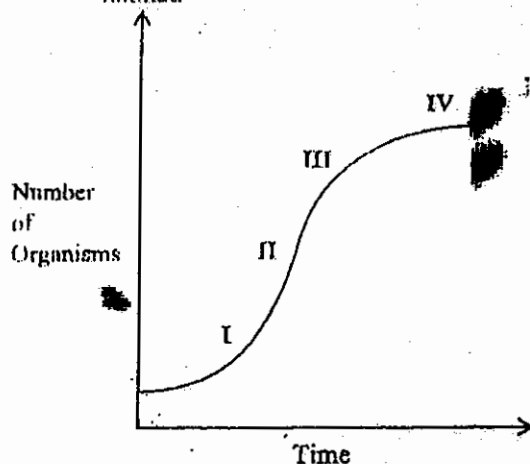
56. Which of the following are appropriate methods for measuring biotic and abiotic factors respectively?

	Biotic factors	Abiotic factors
(A)	Quadrat, sweep net, jar	Thermometer, pH paper, light meter
(B)	Light meter, pH paper, quadrat	Thermometer, jar, sweep net
(C)	Thermometer, sweep net, jar	Light meter, pH paper, quadrat
(D)	Sweep net, jar, thermometer	Light meter, pH paper, quadrat

57. The term 'habitat' refers to

- (A) an area where several organisms live
(B) the physical characteristics of an area
(C) the biological characteristics of an area
(D) the area in which a particular organism lives

Item 58 refers to the figure showing the growth curve of a population colonising a new habitat.



58. Which of the following accounts for phase I?

- (A) There is a small number of reproducing individuals.
(B) There is competition for resources.
(C) There is competition for food.
(D) Death rate exceeds birth rate.

59. Non-renewable resources include

- I. bauxite and gold
II. forests and swamps/wetlands
III. natural gas and oil

- (A) I and II only
(B) I and III only
(C) II and III only
(D) I, II and III

60. Which of the following practices does NOT help to conserve the environment?

- (A) Using natural fertilizers
(B) Reforestation
(C) Crop rotation
(D) Overgrazing

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.